

Compact $U(1)$ under refinement: an explicit total-variation bound in three dimensions, and why it fails in four

Result, 2026-09-27. For the heat-kernel (Villain) $U(1)$ lattice gauge theory on the periodic lattice $(\mathbb{Z}/N)^D$ of spacing a , physical side $L = Na$ and plaquette heat time t , the lattice measure splits exactly into a monopole-free part (the free photon together with the physical flux sectors of the torus) and a gas of lattice monopoles with Coulomb weights (Proposition 1; the duality of Banks, Myerson and Kogut). A bound on the lattice Laplacian then gives:

- $D = 3$ (**Theorem 2**). With $t = \lambda_3 a \leq 1$,

$$\|\mu_a - \mu_a^0\|_{\text{TV}} \leq \frac{2(L/a)^3 e^{-\pi^2/(6\lambda_3 a)}}{1 - e^{-\pi^2/(2\lambda_3 a)}},$$

where μ_a^0 is the monopole-free measure. The monopole free-energy density per physical volume lies between $a^{-3} \cdot 6e^{-2\pi^2/(3\lambda_3 a)}$ (up to a factor tending to one) and $a^{-3} \cdot 2e^{-\pi^2/(6\lambda_3 a)} / (1 - e^{-\pi^2/(2\lambda_3 a)})$. At fixed λ_3 and L everything compact vanishes faster than any power of a ; along trajectories with $\lambda_3 a \simeq c/\log(1/a)$ it stays finite. The flux sectors carry the weight $e^{-2\pi^2|m|^2/(\lambda_3 L)}$, independent of a .

- $D = 4$ (**Theorem 3, and a recorded failure**). With $t = g^2$, the same decomposition holds with flux-sector weight $e^{-2\pi^2|m|^2/g^2}$, independent of both a and L , and the monopole-loop free energy per lattice cell lies between $12e^{-\pi^2/g^2}$ and $8e^{-\pi^2/(8g^2)} / (1 - e^{-3\pi^2/(8g^2)})$, up to factors tending to one. The total-variation bound is then of order $(L/a)^4 e^{-\pi^2/(8g^2)}$, which grows as $a \rightarrow 0$ at fixed g . The criterion that closes the three-dimensional case fails here for an exact reason: the loop density per lattice cell is fixed by g alone. The four-dimensional continuum statement (Driver's renormalized free field) needs observables and a coupling renormalization, which a partition-function bound cannot supply.

This is the first step of the $D = 3$ programme of STATE carried out at the level of the measure, with explicit a , L and coupling, for the smallest compact group. It sharpens Theorem 5 of the series/parallel note (one step, rate $e^{-\pi^2/(8t)}$) into a bound on the whole lattice measure with rate $e^{-\pi^2/(6t)}$, and it puts numbers on the error-budget row of the zero-spacing note. The duality is established (Banks, Myerson and Kogut 1977, metadata) and the continuum limit at fixed coupling is Gross's theorem (Gross 1983, abstract as indexed); the explicit total-variation bound and the flux-sector bookkeeping are elementary consequences, with no novelty claimed.

1. The exact decomposition

Let E , P and C_3 be the links, plaquettes and 3-cells of the periodic hypercubic lattice $(\mathbb{Z}/N)^D$, d the coboundary on cochains, and $\Delta = dd^\top + d^\top d$ the Hodge Laplacian. The heat-kernel measure on link angles $\theta \in [0, 2\pi)^E$ is

$$\mu_a(d\theta) = \frac{1}{Z} \prod_{p \in P} k_t((d\theta)_p) \frac{d\theta}{(2\pi)^{|E|}}, \quad k_t(\phi) = \sum_{n \in \mathbb{Z}} e^{-tn^2/2} e^{in\phi} = \sqrt{\frac{2\pi}{t}} \sum_{n \in \mathbb{Z}} e^{-(\phi + 2\pi n)^2/(2t)}.$$

Write $\mathbb{R}^P = \text{im } d \oplus \text{im } d^\top \oplus \mathcal{H}$ (exact, coexact and harmonic 2-cochains; $\mathcal{H}$ is spanned by the cochains constant on each of the $\binom{D}{2}$ plaquette orientations). For $n \in \mathbb{Z}^P$ let $q = dn \in \mathbb{Z}^{C_3}$ (the monopole charges) and $n_{\mathcal{H}}$ the harmonic projection of n .

Proposition 1 (monopole decomposition). μ_a is a mixture

$$\mu_a = \frac{1}{Z} \sum_{[n] \in \mathbb{Z}^P / d\mathbb{Z}^E} w([n]) \nu_{[n]}, \quad w([n]) = \exp\left\{-\frac{2\pi^2}{t} \left(q^\top (dd^\top)^+ q + |n_{\mathcal{H}}|^2\right)\right\},$$

of probability measures $\nu_{[n]}$ on link angles, where $(dd^\top)^+$ is the pseudo-inverse on $\text{im } d$ and $|\cdot|$ the Euclidean norm on cochains. The classes $[n]$ with $q = 0$ make up the monopole-free measure μ_a^0 .

Proof. Insert the Poisson form of k_t and exchange sum and integral. The substitution $n \mapsto n + d\ell$, $\theta \mapsto \theta - 2\pi\ell$ ($\ell \in \mathbb{Z}^E$) groups the sum over n into classes and extends the angle integral to $\mathbb{R}^E$ modulo 2π times the integer closed 1-cochains, a lattice of full rank in the closed 1-cochains; the remaining integrand depends on θ only through $\omega = d\theta$. So the class $[n]$ contributes a constant times $\int_{\text{im } d} e^{-|\omega + 2\pi n|^2/(2t)} d\omega$. Decompose $n = n_{\text{ex}} + n_{\text{co}} + n_{\mathcal{H}}$ orthogonally. The exact part is absorbed by translating ω , and $|n_{\text{co}}|^2 = (dn)^\top (dd^\top)^+ (dn)$ because $n_{\text{co}} = d^\top (dd^\top)^+ dn$. The remaining Gaussian integral is the same for every class. $\square$

The classes are labelled by $q \in d\mathbb{Z}^P$ together with an integer flux vector $m \in \mathbb{Z}^{\binom{D}{2}}$ through the coordinate 2-tori (the integral cohomology of the torus is free). For fixed q the harmonic part is $n_{\mathcal{H}} = h(m) + y(q)$ with $h(m)$ the constant cochain of flux m and $y(q)$ an offset fixed by q . A cochain of flux $m_{\mu\nu}$ through each $(\mu\nu)$ torus has value $m_{\mu\nu}/N^2$ on each of the N^D plaquettes of that orientation, so

$$|h(m)|^2 = N^{D-4} \sum_{\mu < \nu} m_{\mu\nu}^2.$$

Two elementary facts. (i) Each Fourier mode of Δ on the hypercubic lattice has eigenvalue $\sum_{\mu} 4 \sin^2(k_{\mu}/2) \leq 4D$, so $\|dd^\top\| \leq 4D$ and $q^\top (dd^\top)^+ q \geq |q|^2/(4D)$ on $\text{im } d$. (ii) For y fixed, the theta-type sum $\sum_m e^{-\alpha|h(m)+y|^2}$ lies between $e^{-\alpha|y|^2} \sum_m e^{-\alpha|h(m)|^2}$ (pair m with $-m$) and $\sum_m e^{-\alpha|h(m)|^2}$ (its Fourier coefficients are positive).

The elementary monopole pair or loop. For a single plaquette p , the charge $q = d e_p$ is a nearest monopole pair ($D = 3$) or an elementary monopole loop ($D = 4$), and $q^\top (dd^\top)^+ q = |P_{\text{co}} e_p|^2$. By translation invariance each plaquette has the same projections, so $|P_{\text{ex}} e_p|^2 = \text{rank } d_1/|P|$ and $|P_{\mathcal{H}} e_p|^2 = \binom{D}{2}/|P|$. With $\text{rank } d_1 = |E| - (N^D - 1) - D$:

$$D = 3: |P_{\text{co}} e_p|^2 = \frac{1}{3} \left(1 - \frac{1}{N^3}\right), \quad D = 4: |P_{\text{co}} e_p|^2 = \frac{1}{2} \left(1 - \frac{1}{N^4}\right).$$

For $D = 3$ this agrees with the lattice Green's function: $G(0) - G(e_1) = \frac{1}{6}(1 - N^{-3})$, so a nearest pair has energy $2(G(0) - G(e_1))$.

2. Three dimensions: the theorem

Theorem 2. Let $D = 3$, $N = L/a$, $t = \lambda_3 a \leq 1$, and $R = Z/Z^0$ the ratio of the full partition function to its monopole-free part. Then

$$1 + 6N^3 \exp\left\{-\frac{2\pi^2}{3\lambda_3 a} - \frac{2\pi^2 a^2}{\lambda_3 L^3}\right\} \leq R \leq \exp\left\{\frac{2N^3 e^{-\pi^2/(6\lambda_3 a)}}{1 - e^{-\pi^2/(2\lambda_3 a)}}\right\},$$

and

$$\|\mu_a - \mu_a^0\|_{\text{TV}} \leq 1 - \frac{1}{R} \leq \frac{2(L/a)^3 e^{-\pi^2/(6\lambda_3 a)}}{1 - e^{-\pi^2/(2\lambda_3 a)}}.$$

The monopole-free part is the free photon on the torus with the flux sectors weighted by $\exp\{-2\pi^2|m|^2/(\lambda_3 L)\}$.

Proof. The flux weight is $|h(m)|^2 = N^{-1}|m|^2$ and $tN = \lambda_3 L$. Upper bound: by fact (ii) the sum over m at fixed q is at most its value at $q = 0$, so $R \leq \sum_{q \in d\mathbb{Z}^P} e^{-(2\pi^2/t)q^\top (dd^\top)^+ q}$. By fact (i) with $4D = 12$ each term is at most $e^{-(\pi^2/6t)|q|^2}$. Enlarge the sum to all of $\mathbb{Z}^{C_3}$, $|C_3| = N^3$, and use $\sum_{k \in \mathbb{Z}} e^{-\alpha k^2} \leq 1 + 2e^{-\alpha}/(1 - e^{-3\alpha})$ (from $k^2 - 1 \geq 3(k - 1)$) with $\alpha = \pi^2/(6t)$, then $\log(1 + x) \leq x$. Lower bound: keep $q = 0$ and the $6N^3$ nearest pairs $q = \pm d e_p$ ($3N^3$ plaquettes, two signs). Each pair has Coulomb energy at most $\frac{1}{3}$ and harmonic offset $|y|^2 = |P_{\mathcal{H}} e_p|^2 = N^{-3}$,

and fact (ii) gives the stated factor. Total variation: μ_a is the mixture $R^{-1}\mu_a^0 + (1 - R^{-1})\mu_a^{\text{rest}}$, so $\|\mu_a - \mu_a^0\|_{\text{TV}} \leq 1 - R^{-1} \leq \log R$. $\square$

Reading. In a box of fixed physical side, the lattice measure is within total variation $O((L/a)^3 e^{-\pi^2/(6\lambda_3 a)})$ of the monopole-free measure. At fixed λ_3 this tends to zero faster than any power of a . By the summable-error criterion of the refinement note, Proposition 4, the continuum question for the compact theory then reduces to that for the free photon with flux sectors, which is Gaussian; Gross's theorem is the continuum statement. The monopole free-energy density $L^{-3} \log R$ lies between the two bounds in the summary. It stays finite exactly when $e^{-c/(\lambda_3 a)} \sim a^3$, that is along $\lambda_3 a \simeq c/(3 \log(1/a))$ with c between $\pi^2/6$ and $2\pi^2/3$: the trajectories on which a Debye mass survives (Göpfert–Mack 1982, abstract as indexed). The exponents are bounds; the monopole self-energy of the infinite lattice, $2\pi^2 G(0) \approx 4.99$ with $G(0) \approx 0.2527$, lies between them.

3. Four dimensions: the same bound, and why it fails

Theorem 3. Let $D = 4$, $N = L/a$, $t = g^2$ with $g^2 \leq 1$. Then

$$1 + 12N^4 \exp\left\{-\frac{\pi^2}{g^2} - \frac{2\pi^2}{g^2 N^4}\right\} \leq R \leq \exp\left\{\frac{8N^4 e^{-\pi^2/(8g^2)}}{1 - e^{-3\pi^2/(8g^2)}}\right\},$$

$\|\mu_a - \mu_a^0\|_{\text{TV}} \leq \log R$, and the flux sectors carry the weight $\exp\{-(2\pi^2/g^2) \sum_{\mu < \nu} m_{\mu\nu}^2\}$.

Proof. As for Theorem 2, with $4D = 16$, $|C_3| = 4N^4$, $|h(m)|^2 = |m|^2$, $6N^4$ plaquettes and two signs for the elementary loops, loop energy at most $\frac{1}{2}$, and harmonic offset N^{-4} . $\square$

The recorded failure. In four dimensions the heat time is the coupling itself, so at fixed g the monopole-loop free energy per lattice cell is a fixed number between the two bounds, and the total-variation bound grows like $(L/a)^4$. The three-dimensional route, closeness of the whole lattice measure to its Gaussian part, is unavailable in four dimensions for an exact reason, and the free-energy density it controls is a vacuum constant. The known continuum statement is different in kind: the compact theory converges on its current sector to a renormalized free electromagnetic field (Driver 1987, abstract as indexed), with the monopole loops renormalizing the charge. A four-dimensional theorem in the insertion language therefore has to control observables and a coupling shift per step, as Hypothesis P(α) does, rather than the measure. This is the four-dimensional row of the error budget in the zero-spacing note, now with the density per cell made explicit: between $12e^{-\pi^2/g^2}$ and $8e^{-\pi^2/(8g^2)}$ up to factors tending to one.

4. Consequence for STATE

Item 1 of STATE gains a measure-level theorem for $U(1)$ in $D = 3$ with explicit a , L , λ_3 (Theorem 2), and a recorded failure of the same route in $D = 4$ with its reason (Theorem 3). The non-abelian step that corresponds to Theorem 2 needs a replacement for the Poisson decomposition: for $SU(2)$ the exact kernel (11) of the series/parallel note has image terms of both signs, so the mixture of Proposition 1 is not available and the Laplace-remainder route of its Proposition 7 remains the path.